MIL-W-50932 (MU)
20 July 1971
SUPERSEDING
(See 6.5

MILITARY SPECIFICATION

WARHEAD SECTION, 762MM ROCKET, TRAINING: M42, 23

OGIVE FOR

1. SCOPE

1.1 This specification covers one type of ogive for use with Warhead Section, 762MM Rocket, Training: M42.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	-	Plating Cadmium (Electrodeposited)
QQ-P-325	-	Zinc Plating (Electrodeposited)

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons General Specification for
MIL-C-5541	-	Chemical Treatment of Aluminum and Aluminum Alloys
MIL-I-6866	-	Inspection, Penetrant, Method of
MIL-I-45607	-	Inspection Equipment, Acquisition Maintenance and Disposition of

FSC: 1340

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MIL-T-5021	-	Test, Aircraft Welding Operators Certification of
MIL-W-8604	-	Welding of Aluminum Alloys Process for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8863699	-	Ogive, Rocket
8887961	-	Carton, Packing, Ammunition for Ogive, Rocket
8887965	-	Box, Packing, Ammunition, for Ogive, Rocket

PUBLICATIONS

ARMY

ET-8863699	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8 - 69 - Standard Method of Tension Testing of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (Dwg.) F-8863699, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Welding.-The weld of each ogive shall be such as to produce satisfactory joined parts. All welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of good fusion, penetration and soundness of deposit representative of good welding practice. The process shall be such as to prevent warpage beyond dimensional requirements. The satisfactoriness of the weld shall be based upon compliance with 3.3.1, 3.3.2, 3.3.3, 3.3.4, 3.3.5 and 3.4.2.

3.3.1 Electrodes.-Electrodes shall be of a coated type, except as specified in 3.3.2. The coating shall be uniform and without bare spots, except those used to introduce current to the electrode. All types of electrodes to be used shall be approved by the Government.

3.3.2 Process.-Gas shielded arc-welding or other automatic arc-welding in which bare or lightly coated wire is used with the arc bathed in a reducing gas or other atmosphere excluding media, may be used.

3.3.3 Qualification.

3.3.3.1 Welding operators.-Welding operators shall be qualified in accordance with MIL-T-5021.

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3.3.3.2 Welding equipment.-Welding equipment shall be qualified in accordance with MIL-W-8604.

3.3.3.3 All welding shall be accomplished in conformance with the process used to qualify the first article sample.

3.3.4 Acceptable weld imperfections permitted in Ogive.

3.3.4.1 Sag.-Sag of the material in the area adjacent to the weld shall not be more than .010 inch below the surface of the joined part.

3.3.4.2 Porosity.-Porosity or other discontinuities shall not extend as surface cracks in excess of those defined in 3.3.5 as determined by examination in accordance with 4.4.3, except that microporsity as evidence by partial penetration into skins by penetrant, shall be acceptable.

3.3.4.3 Blow-outs.-Blow-outs may be repaired by hand welding operations, provided they are no greater than one quarter (1/4) inch in width and one (1) inch in length and not more than three (3) such areas may be repaired in any one skin.

3.3.5 Surface cracks.-Surface cracks shall not be in excess of the following:

3.3.5.1 There shall be no more than one (1) surface crack 1/2 inch long in each one (1) inch of weld.

3.3.5.2 There shall be no more than a total of two (2) inches of surface cracks in each weld.

3.4 Mechanical properties.

3.4.1 Component parts.-The aluminum used in the fabrication of the component parts shall comply with the minimum (min) properties specified on the applicable drawing and the heat treat condition required, when tested as specified in 4.2.2.1 or 4.3.3.2, as applicable.

3.4.2 Welds (aluminum).-The mechanical properties of the welds shall not be less than 30,000 pounds per square inch (psi) tensile strength when tested as specified in 4.2.2.2. Unless otherwise directed by the Contracting Officer, this test shall be performed on the first article sample only.

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3.5 Functioning Test (Assembly) of Ogive.-The Ogive shall assemble easily and with no use of force and in accordance with the intent of the design when tested as specified in 4.4.1.

3.6 Workmanship.-All parts and assemblies shall be fabricated, finished, and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Surface coatings shall be continuous except for a few small scratches not exposing base material. All required markings shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.8 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.

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c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the cerfitying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of one (1) complete Ogive and three (3) complete sets of parts (i.e. three (3) each of every component and every sub-assembly). All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be submitted by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed.

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4.2.2.1 Mechanical properties of Component parts.-Four (4) specimens complying with ASTM-E8-69 (part 6), shall be taken from one (1) ogive for tensile and yield strength tests. The four (4) specimens shall be taken as follows: two (2) shall be taken parallel to the longitudinal axis and two (2) normal to the axis of the skin. The test shall be in accordance with 4.4.4.

4.2.2.2 Tensile properties of skin welds.-Two (2) specimens, complying with the type specified in ASTM E8-69 shall be taken transverse to the welded joint from the skin tested and two (2) specimens shall be taken parallel to longitudinal axis of the welded joint of the skin. Each weld shall be tested for conformance to requirement 3.4.2 using Method specified in ASTM E8-69 (part 6).

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of

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an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cone tip assembly (See Dwg. 8798096).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Pitch diameter of thread	Gage	01001
102.	Minor diameter of thread	Gage	01002
103.	Location of tip insert	Gage	01003
Minor	AQL 1.50 Percent		
201.	Casting chipped or cracked	Visual	01004
202.	Insert loose	Visual-Manual	01005
203.	Evidence of poor workmanship (See 3.6)	Visual	01006

4.3.2.2 Nose centering ring assembly (See Dwg. 8798071).-

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Categories	Defects	Method of Inspection	Code No.
Critical and Major: None defined			
Minor	AQL 1.50 Percent		
201.	Length, max	Gage	02001
202.	Concentricity of tube with forward end	Gage	02002
203.	Pitch diameter of tapped holes	Gage	02003
204.	Evidence of poor workmanship (See 3.6)	Visual	02004

4.3.2.3 Center tube assembly (See Dwg 8863753)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major	AQL 0.40 Percent		
101.	Pitch diameter of thread	Gage	03001
102.	Major diameter of thread	Gage	03002
Minor	AQL 1.50 Percent		
201.	Alignment of tapped hole with inner tube hole	Visual	03003
202.	Spring pin protruding	Visual	03004
203.	Evidence of poor workmanship (See 3.6)	Visual	03005

4.3.2.4 Skin, nose (See Dwg 8798084)

Categories	Defects	Method of Inspection	Code No.
Critical and Major: None defined			
Minor	AQL 2.50 Percent		
201.	Diameter, small end	Gage	04001
202.	Diameter, large end	Gage	04002
203.	Length	Gage	04003
204.	Thickness	Gage	04004
205.	Evidence of poor workmanship (See 3.6)	Visual	04005

4.3.2.5 Ogive, Rocket (See Dwg 8863699)

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Part missing	Visual	05001
Minor	AQL 2.50 Percent		
201.	Ring, lock, improperly assembled	Visual	05002
202.	Block, plug-in, damaged	Visual	05003
203.	Lubricant or primer missing	Visual	05004
204.	Markings misleading or unidentifiable	Visual	05005
205.	Evidence of poor workmanship (See 3.6)	Visual	05006

4.3.2.6 Carton, Packing, Ammunition, for Ogive Rocket (See dwg 8887961)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Ogive missing	Visual	06001
Minor	AQL Percent		
201.	Container damaged	Visual	06002
202.	Marking misleading or unidentifiable	Visual	06003
203.	Evidence of poor workmanship (See 3.6)	Visual	06004

4.3.2.7 Box, packing, ammunition, for Ogive, Rocket (See dwg. 8887965).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL Percent		
201.	Box damaged	Visual	07001
202.	Markings misleading or unidentifiable	Visual	07002
203.	Evidence of poor workmanship (See 3.6)	Visual	07003

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4.3.3 Testing

4.3.3.1 Functioning test (assembly) of Ogive (See 3.5) Major defect, Code No. 08001.-Each Ogive, at the time of submission shall be tested for satisfactory functioning. Any ogive failing to meet this requirement shall be classed defective and removed from the lot. This is a non-destructive test. Parts so tested may be returned to the lot.

4.3.3.2 Salt Spray for corrosive resistance (See applicable drawing) Major defect, Code No. 09001.-Sample coupons or specimens representative of the material and finish of the finished parts shall be prepared in the quantity and manner specified in QQ-Z-325, QQ-P-416 or MIL-C-5541, as applicable. If any part fails to comply with the corrosive resistance requirement action shall be taken as specified in applicable specification and the batch or lot represented by the failing sample shall be rejected.

4.3.3.3 Penetrant examination of welds (See 3.3.4.2) Major Defects Code No. 10001.-Each weld shall be examined as specified in 4.4.3.

4.3.3.4 Mechanical properties (See Applicable Dwg.) Major Defect, Code No. 11001.-Test specimens of the same material and heat treatment as that used in the ogive assembly shall be tested in accordance with ASTM E8 (Part 6). If any part fails to comply with the applicable requirements, the lot represented by the failing sample shall be rejected.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8863699 and Equipment Lists (EL's) referenced theron identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

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4.4 Test methods and procedures.

4.4.1 Functioning test (assembly) of Ogive,-The ogive, shall be assembled to a fixture as described in the equipment list. The three (3) set screws shall be loosened and the cone tip turned in a counter-clockwise direction, using a suitable tool. The ogive, shall be pulled out until the plug-in block s are free of the fixture. The ogive, shall then be re-assembled and the set-screws tightened.

4.4.2 Salt spray for corrosive resistance.-Salt spray test shall be performed in accordance with MIL-C-5541, QQ-P-416 or QQ-P-325.

4.4.3 Penetrant examination of skin welds.-The Skin weld shall be examined as specified in MIL-I-6866. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Mechanical properties.-The ogive specimens shall be subjected to the mechanical properties testing in accordance with ASTM E8-69 (Part 6). All data shall be recorded.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.-The ogive shall be packed in accordance with Drawings 8887961 and 8887965.

5.2 Marking.-Marking shall be in accordance with Drawings 8887961 and 8887965.

5.3 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

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6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

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6.5 Supersession data.-This military specification includes the requirements of and supersedes, Purchase specification AEI-139 (Rev 1), dated 1 Nov. 1964.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1340-A

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